

3D Trigonometry — Solutions

Warm-up

Question 1:

The base of the box is a rectangle with dimensions 8 cm by 6 cm. The diagonal of the base, d_{base} , forms the hypotenuse of a right-angled triangle with the sides of the rectangle.

$$\begin{aligned}\text{By Pythagoras' Theorem: } d_{base}^2 &= 8^2 + 6^2 \\ d_{base}^2 &= 64 + 36 \\ d_{base}^2 &= 100 \\ d_{base} &= \sqrt{100} \\ d_{base} &= 10 \text{ cm}\end{aligned}$$

Question 2:

Let the side length of the cube be s . The internal diagonal connects two opposite vertices of the cube. First, we find the diagonal of the base, d_{base} .

$$\begin{aligned}\text{Base diagonal: } d_{base}^2 &= s^2 + s^2 \\ d_{base}^2 &= 2s^2\end{aligned}$$

Now, we use the base diagonal and the vertical edge s to find the internal diagonal D .

$$\begin{aligned}\text{Internal diagonal: } D^2 &= d_{base}^2 + s^2 \\ D^2 &= 2s^2 + s^2 \\ D^2 &= 3s^2 \\ D &= s\sqrt{3}\end{aligned}$$

Question 3:

Let OT be the vertical mast of height h . Point A is on the ground. The angle of elevation from A to T is 25° . This forms a right-angled triangle $\triangle OTA$ where $\angle OTA = 90^\circ$.

$$\begin{aligned}\tan 25^\circ &= \frac{\text{opposite}}{\text{adjacent}} \\ \tan 25^\circ &= \frac{h}{OA} \\ OA &= \frac{h}{\tan 25^\circ} \\ OA &= h \cot 25^\circ\end{aligned}$$

Question 4:

Given $OA = h \cot 40^\circ$ and $OB = h \cot 50^\circ$, and $\angle AOB = 90^\circ$. The points A, O, B form a right-angled triangle $\triangle AOB$ on the ground.

By Pythagoras' Theorem: $AB^2 = OA^2 + OB^2$

$$AB^2 = (h \cot 40^\circ)^2 + (h \cot 50^\circ)^2$$

$$AB^2 = h^2(\cot^2 40^\circ + \cot^2 50^\circ)$$

Question 5:

In $\triangle PQR$, we have $PQ = 10$, $QR = 12$ and $\angle PQR = 60^\circ$. We use the Cosine Rule to find PR .

$$PR^2 = PQ^2 + QR^2 - 2(PQ)(QR) \cos(\angle PQR)$$

$$PR^2 = 10^2 + 12^2 - 2(10)(12) \cos 60^\circ$$

$$PR^2 = 100 + 144 - 240 \left(\frac{1}{2}\right)$$

$$PR^2 = 244 - 120$$

$$PR^2 = 124$$

$$PR = \sqrt{124} = 2\sqrt{31} \approx 11.14$$

Question 6:

Point P is due North of O and point Q is due East of O . The directions North and East are perpendicular to each other.

$$\angle POQ = 90^\circ$$

Skill-building

Question 1:

Let $OT = h$ be the height of the tower. A is due West of O and B is due South of O . Thus, $\angle AOB = 90^\circ$. In $\triangle OTA$ (right-angled at O):

$$\tan 35^\circ = \frac{h}{OA} \implies OA = h \cot 35^\circ$$

In $\triangle OTB$ (right-angled at O):

$$\tan 45^\circ = \frac{h}{OB} \implies OB = h \cot 45^\circ = h(1) = h$$

Now, consider $\triangle AOB$ on the ground. Since $\angle AOB = 90^\circ$, we use Pythagoras' Theorem:

$$AB^2 = OA^2 + OB^2$$

$$AB^2 = (h \cot 35^\circ)^2 + h^2$$

$$AB^2 = h^2 \cot^2 35^\circ + h^2$$

$$AB^2 = h^2(\cot^2 35^\circ + 1)$$

Question 2:

In the rectangular prism, $AB = 10$ and $BC = 15$. The base is $ABCD$. The diagonal of the base AC is found using Pythagoras' Theorem:

$$\begin{aligned}
AC^2 &= AB^2 + BC^2 \\
AC^2 &= 10^2 + 15^2 \\
AC^2 &= 100 + 225 \\
AC^2 &= 325 \\
AC &= \sqrt{325}
\end{aligned}$$

Let AE be the height of the prism. The angle of elevation from C to E is α . In the right-angled triangle $\triangle ACE$ (where E is vertically above A):

$$\begin{aligned}
\tan \alpha &= \frac{\text{opposite}}{\text{adjacent}} \\
\tan \alpha &= \frac{AE}{AC} \\
\tan \alpha &= \frac{AE}{\sqrt{325}}
\end{aligned}$$

Question 3:

P is due East of X , and X, Y are on a North-South line, so $\angle PXY = 90^\circ$. In $\triangle BPX$ (right-angled at P):

$$\tan 40^\circ = \frac{h}{XP} \implies XP = h \cot 40^\circ$$

In $\triangle BPY$ (right-angled at P):

$$\tan 30^\circ = \frac{h}{YP} \implies YP = h \cot 30^\circ$$

In $\triangle XPY$ (right-angled at X):

$$\begin{aligned}
YP^2 &= XP^2 + XY^2 \\
(h \cot 30^\circ)^2 &= (h \cot 40^\circ)^2 + 200^2 \\
h^2 \cot^2 30^\circ - h^2 \cot^2 40^\circ &= 200^2 \\
h^2(\cot^2 30^\circ - \cot^2 40^\circ) &= 40000 \\
h^2 &= \frac{40000}{\cot^2 30^\circ - \cot^2 40^\circ} \\
h &= \sqrt{\frac{40000}{3.333^2 - 1.191^2}} \\
&\approx \sqrt{4117.5} \approx 64.17 \text{ m}
\end{aligned}$$

Question 4:

Let O be the base of the flagpole. $OA = 50$ m, bearing 040°T . $OB = 80$ m, bearing 160°T . The angle $\angle AOB$ is the difference between the bearings:

$$\angle AOB = 160^\circ - 40^\circ = 120^\circ$$

To find AB , we use the Cosine Rule in $\triangle AOB$:

$$\begin{aligned}
AB^2 &= OA^2 + OB^2 - 2(OA)(OB) \cos 120^\circ \\
AB^2 &= 50^2 + 80^2 - 2(50)(80)(-0.5) \\
AB^2 &= 2500 + 6400 + 4000 \\
AB^2 &= 12900 \\
AB &= \sqrt{12900} \approx 113.58 \text{ m}
\end{aligned}$$

Given $h = 20$, the angle of elevation α from A to the top of the pole T is found from $\triangle OTA$ (right-angled at O):

$$\begin{aligned}
\tan \alpha &= \frac{h}{OA} \\
\tan \alpha &= \frac{20}{50} = 0.4 \\
\alpha &= \tan^{-1}(0.4) \approx 21.8^\circ
\end{aligned}$$

Question 5:

Let PQ be the vertical mast of height h . A is at bearing 180°T (South since the mast is North of A) and B is at bearing 270°T (West since the mast is east of B). Thus, $\angle AOB = 90^\circ$ where O is the base of the mast. In $\triangle OPA$ (right-angled at O):

$$\tan 20^\circ = \frac{h}{OA} \implies OA = h \cot 20^\circ$$

In $\triangle OPB$ (right-angled at O):

$$\tan 30^\circ = \frac{h}{OB} \implies OB = h \cot 30^\circ$$

To find the bearing of B from A , we need $\angle OAB$ in $\triangle AOB$ (right-angled at O):

$$\begin{aligned}
\tan(\angle OAB) &= \frac{OB}{OA} = \frac{h \cot 30^\circ}{h \cot 20^\circ} = \frac{\cot 30^\circ}{\cot 20^\circ} \\
\angle OAB &= \tan^{-1} \left(\frac{1.732}{2.747} \right) \approx 32.2^\circ
\end{aligned}$$

Since A is due South of O , the line AO is the North direction from A . The bearing of B from A is $360^\circ - \angle OAB$ if B is East of O . Thus, the bearing is 327.7°T .

Question 6:

The person walks 500 m up a path inclined at 15° to the horizontal. This forms a right-angled triangle where the hypotenuse is 500 m and the angle is 15° .

$$\begin{aligned}
\sin 15^\circ &= \frac{\text{vertical rise}}{\text{hypotenuse}} \\
\text{Vertical rise} &= 500 \sin 15^\circ \\
\text{Vertical rise} &= 500(0.2588) \approx 129.4 \text{ m}
\end{aligned}$$

Question 7:

Let OT be the tower of height h . A is a point such that the bearing of T from A is 030°T .

B is 100 m due East of A . The bearing of T from B is 330°T . In $\triangle AOB$ (where O is the base of the tower): $\angle OAB = 30^\circ$ (since A is at 030°T from O is not the case, the bearing of T from A is 030°T , and AB is East, which is 090°T). Thus, $\angle OAB = 90^\circ - 30^\circ = 60^\circ$. $\angle OBA = 360^\circ - 330^\circ + 90^\circ$ is not correct. The bearing of T from B is 330°T . The line BA is West (270°T). Thus, $\angle OBA = 330^\circ - 270^\circ = 60^\circ$. Since $\angle OAB = 60^\circ$ and $\angle OBA = 60^\circ$, $\triangle AOB$ is equilateral.

$$OA = AB = 100 \text{ m}$$

Now, using the angle of elevation from A to T (25°):

$$\tan 25^\circ = \frac{h}{OA} \implies h = 100 \tan 25^\circ \approx 46.6 \text{ m}$$

Question 8:

Let the rectangular base be 12 cm by 8 cm. Let O be the center of the base. The diagonal of the base is $d_{base} = \sqrt{12^2 + 8^2} = \sqrt{144 + 64} = \sqrt{208} = 4\sqrt{13}$ cm. The distance from the center O to a corner is $r = \frac{1}{2}d_{base} = 2\sqrt{13}$ cm. Let V be the apex. The slanted edge is 15 cm. In the right-angled triangle formed by the height VO , the slanted edge VA , and the distance OA :

$$\begin{aligned} \cos \theta &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{OA}{VA} \\ \cos \theta &= \frac{2\sqrt{13}}{15} \\ \theta &= \cos^{-1} \left(\frac{2\sqrt{13}}{15} \right) \approx 55.1^\circ \end{aligned}$$

Question 9:

Given $OA = h \tan 70^\circ$, $OB = h \tan 60^\circ$, and $AB = 100$ with $\angle AOB = 45^\circ$. We apply the Cosine Rule to $\triangle AOB$:

$$\begin{aligned} AB^2 &= OA^2 + OB^2 - 2(OA)(OB) \cos(\angle AOB) \\ 100^2 &= (h \tan 70^\circ)^2 + (h \tan 60^\circ)^2 - 2(h \tan 70^\circ)(h \tan 60^\circ) \cos 45^\circ \\ 10000 &= h^2(\tan^2 70^\circ + \tan^2 60^\circ - 2 \tan 70^\circ \tan 60^\circ \cos 45^\circ) \\ h^2 &= \frac{10000}{\tan^2 70^\circ + \tan^2 60^\circ - 2 \tan 70^\circ \tan 60^\circ \cos 45^\circ} \\ h &= \frac{100}{\sqrt{\tan^2 70^\circ + \tan^2 60^\circ - 2 \tan 70^\circ \tan 60^\circ \cos 45^\circ}} \end{aligned}$$

Question 10:

Let h be the height of the tower OT . Let A and B be two points on the ground. The angles of elevation are α and β respectively. In $\triangle OTA$ (right-angled at O):

$$\tan \alpha = \frac{h}{OA} \implies OA = h \cot \alpha$$

In $\triangle OTB$ (right-angled at O):

$$\tan \beta = \frac{h}{OB} \implies OB = h \cot \beta$$

In $\triangle AOB$, let $\angle AOB = \theta$ and $AB = d$. Using the Cosine Rule:

$$\begin{aligned} d^2 &= OA^2 + OB^2 - 2(OA)(OB) \cos \theta \\ d^2 &= (h \cot \alpha)^2 + (h \cot \beta)^2 - 2(h \cot \alpha)(h \cot \beta) \cos \theta \\ d^2 &= h^2 \cot^2 \alpha + h^2 \cot^2 \beta - 2h^2 \cot \alpha \cot \beta \cos \theta \end{aligned}$$

Easier Exam Questions

Question 1:

$$\begin{aligned} MG &= EF - KA \\ &= 11 - 3 \\ &= 8 \\ LG &= AB - LF \\ &= 18 - 3 \\ &= 15 \\ ML^2 &= MG^2 + LG^2 \text{ (Pythagoras Theorem)} \\ &= 8^2 + 15^2 \\ ML &= 17 \\ h &= KM \\ \tan(30) &= \frac{KM}{ML} \\ \sqrt{3} &= \frac{h}{17} \\ h &= 17\sqrt{3} \text{ cm} \\ &\text{Option A} \end{aligned}$$

Question 2:

Given a square pyramid with height $h = 16$ cm and base edge $a = 24$ cm. Let O be the apex and D be a vertex of the base. Let F be the midpoint of BC . (i) OF is the slant height of the face OBC . In $\triangle OFD$ (where F is the midpoint of BC and D is the center of the base):

$$\begin{aligned} OF^2 &= h^2 + (a/2)^2 \\ OF^2 &= 16^2 + 12^2 \\ OF^2 &= 256 + 144 = 400 \\ OF &= 20 \text{ cm} \end{aligned}$$

(ii) The angle between the plane OBC and the base $ABCD$ is $\angle OFD$ (where D is the center of the base):

$$\tan(\angle OFD) = \frac{h}{a/2} = \frac{16}{12} = \frac{4}{3}$$

$$\angle OFD = \tan^{-1}(4/3) \approx 53.1^\circ$$

(iii) DE is the distance from the center to a corner. $DE = \frac{1}{2}\text{diagonal of base} = \frac{1}{2}(24\sqrt{2}) = 12\sqrt{2}$ cm. The angle between the edge OD and the base is $\angle ODE$ (where E is the center of the base):

$$\tan(\angle ODE) = \frac{h}{DE} = \frac{16}{12\sqrt{2}} = \frac{4}{3\sqrt{2}}$$

$$\angle ODE = \tan^{-1}\left(\frac{4}{3\sqrt{2}}\right) \approx 35.3^\circ$$

Question 3:

A weather balloon is h metres above point P . Observer A is due south of the balloon, and observer B is due east of the balloon. $AB = 150$ m. (i) In $\triangle BPA$ (right-angled at P):

$$\tan 58^\circ = \frac{h}{PA} \implies PA = h \cot 58^\circ$$

In $\triangle BPB$ (right-angled at P):

$$\tan 67^\circ = \frac{h}{PB} \implies PB = h \cot 67^\circ$$

(ii) Since A is south and B is east, $\angle APB = 90^\circ$. By Pythagoras' Theorem in $\triangle APB$:

$$AB^2 = PA^2 + PB^2$$

$$150^2 = (h \cot 58^\circ)^2 + (h \cot 67^\circ)^2$$

$$150^2 = h^2(\cot^2 58^\circ + \cot^2 67^\circ)$$

$$h^2 = \frac{150^2}{\cot^2 58^\circ + \cot^2 67^\circ}$$

(iii) Calculating h :

$$\cot 58^\circ \approx 0.6249, \quad \cot 67^\circ \approx 0.4245$$

$$\cot^2 58^\circ \approx 0.3905, \quad \cot^2 67^\circ \approx 0.1802$$

$$h^2 = \frac{22500}{0.3905 + 0.1802} = \frac{22500}{0.5707} \approx 39425.2$$

$$h \approx 198.6 \text{ m}$$

Question 4:

If B is at bearing 055°T from Q , and A is due east (090°T), the angle is 35° . If B is at bearing 055°T from Q , but A is in a different position, let's check. If A is due east of Q , then $\angle AQB = 90 - 55 = 35^\circ$. If the bearing of the tower Q from B is 055°T , then the bearing of B from Q is $55 + 180 = 235^\circ\text{T}$. The angle $\angle AQB$ is the difference between these bearings, so $\angle AQB = 235 - 90 = 145^\circ$.

$$\angle AQB = 235^\circ - 90^\circ = 145^\circ$$

(b) In $\triangle APQ$ (right-angled at Q):

$$\tan 15^\circ = \frac{h}{AQ} \implies AQ = h \cot 15^\circ$$

(c) In $\triangle BPQ$ (right-angled at Q):

$$\tan 12^\circ = \frac{h}{BQ} \implies BQ = h \cot 12^\circ$$

Harder Exam Questions

Question 1:

In the rectangular pyramid, the apex O is the center of the sphere. The perpendicular from O to the base meets the diagonal DB at its midpoint M . $OM = x$ is the height of the pyramid. DB is the diagonal of the rectangular base $ABCD$. Since O is the center of the sphere and D, B are on the surface, $OD = OB = r$ (radius of the sphere). In $\triangle OMB$ (right-angled at M):

$$MB^2 = OB^2 - OM^2$$

$$MB^2 = r^2 - x^2$$

$$MB = \sqrt{r^2 - x^2}$$

Since M is the midpoint of DB , $DB = 2MB$:

$$DB = 2\sqrt{r^2 - x^2}$$

Question 2:

Let h be the height of the tower. P is due east of the tower O , and Q is due south of the tower O . Thus, $\angle POQ = 90^\circ$. In $\triangle OTP$ (right-angled at O):

$$\tan 45^\circ = \frac{h}{OP} \implies OP = h \cot 45^\circ = h(1) = h$$

In $\triangle OTQ$ (right-angled at O):

$$\tan 30^\circ = \frac{h}{OQ} \implies OQ = h \cot 30^\circ = h\sqrt{3}$$

In $\triangle POQ$ (right-angled at O):

$$PQ^2 = OP^2 + OQ^2$$

$$40^2 = h^2 + (h\sqrt{3})^2$$

$$1600 = h^2 + 3h^2$$

$$1600 = 4h^2$$

$$h^2 = 400$$

$$h = 20 \text{ m}$$

Question 3:

Let H be the point vertically above C . The angles of depression from H to A and B are 60° and 30° respectively. Let h be the height of the helicopter. Then HA and HB are the lines of sight. The angles of elevation from A and B to H are also 60° and 30° . In $\triangle HCA$ (right-angled at C):

$$\tan 60^\circ = \frac{h}{CA} \implies CA = h \cot 60^\circ$$

In $\triangle HCB$ (right-angled at C):

$$\tan 30^\circ = \frac{h}{CB} \implies CB = h \cot 30^\circ$$

Given the bearings of A and B from C are 250° and 130° respectively, the angle $\angle ACB$ is the difference between the bearings:

$$\angle ACB = 250^\circ - 130^\circ = 120^\circ$$

Using the Cosine Rule in $\triangle ACB$:

$$\begin{aligned} AB^2 &= CA^2 + CB^2 - 2(CA)(CB) \cos 120^\circ \\ (100\sqrt{13})^2 &= (h \cot 60^\circ)^2 + (h \cot 30^\circ)^2 - 2(h \cot 60^\circ)(h \cot 30^\circ)(-0.5) \\ 130000 &= h^2(\cot^2 60^\circ + \cot^2 30^\circ + \cot 60^\circ \cot 30^\circ) \\ \cot 60^\circ &= 1/\sqrt{3}, \quad \cot 30^\circ = \sqrt{3} \\ 130000 &= h^2(1/3 + 3 + 1) = h^2(13/3) \\ h^2 &= \frac{130000 \times 3}{13} = 30000 \\ h &= 100\sqrt{3} \end{aligned}$$

Now find CA :

$$CA = h \cot 60^\circ = 100\sqrt{3} \times \frac{1}{\sqrt{3}} = 100 \text{ km}$$

Question 4:

Let OT be the tower of height h . A is due south of O . B is 50 m east of A . Thus, $\angle OAB = 90^\circ$.

(a) In $\triangle OTA$ (right-angled at O):

$$\tan 45^\circ = \frac{h}{OA} \implies OA = h \cot 45^\circ = h(1) = h$$

In $\triangle OTB$ (right-angled at O):

$$\tan 30^\circ = \frac{h}{OB} \implies OB = h \cot 30^\circ = h\sqrt{3}$$

(b) In $\triangle OAB$ (right-angled at O):

$$\begin{aligned}
OB^2 &= OA^2 + AB^2 \\
(h\sqrt{3})^2 &= h^2 + 50^2 \\
3h^2 &= h^2 + 2500 \\
2h^2 &= 2500 \\
h^2 &= 1250 \\
h &= \sqrt{1250} = 25\sqrt{2} \text{ m}
\end{aligned}$$

(c) To find the bearing of B from O , we find $\angle AOB$ in $\triangle OAB$:

$$\begin{aligned}
\tan(\angle AOB) &= \frac{AB}{OA} = \frac{50}{25\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \\
\angle AOB &= \tan^{-1}(\sqrt{2}) \approx 54.735^\circ
\end{aligned}$$

Since A is due south of O , the bearing of A is 180° . The bearing of B from O is $180^\circ - 54.735^\circ = 125.265^\circ$. Converting to degrees and minutes: $0.265 \times 60 = 15.9 \approx 16'$. Thus, the bearing is $125^\circ 16'T$.

Question 5:

Let OT be the lighthouse of height h . A is due south of O . B is due east of A . Thus, $\angle OAB = 90^\circ$.

(i) In $\triangle OTA$ (right-angled at O):

$$\tan 68^\circ = \frac{h}{OA} \implies OA = h \cot 68^\circ = h \tan(90^\circ - 68^\circ) = h \tan 22^\circ$$

(ii) In $\triangle OTB$ (right-angled at O):

$$\tan 43^\circ = \frac{h}{OB} \implies OB = h \cot 43^\circ = h \tan(90^\circ - 43^\circ) = h \tan 47^\circ$$

In $\triangle OAB$ (right-angled at O):

$$\begin{aligned}
AB^2 + OA^2 &= OB^2 \\
1500^2 + (h \tan 22^\circ)^2 &= (h \tan 47^\circ)^2 \\
1500^2 &= h^2(\tan^2 47^\circ - \tan^2 22^\circ) \\
h^2 &= \frac{1500^2}{\tan^2 47^\circ - \tan^2 22^\circ} \\
h &= \frac{1500}{\sqrt{\tan^2 47^\circ - \tan^2 22^\circ}}
\end{aligned}$$

Question 6:

Let OT be the tower of height h . A is a point on the ground. The bearing of O from A is 053° . The bearing of O from B is 342° . $AB = 230$ m. In $\triangle AOB$: $\angle OAB = 53^\circ$ (relative to

North). Since the road is straight, we assume the road is the line AB . Let's assume the road is East-West. If the road is North-South, the bearing of O from A is 053° and the bearing of O from B is 342° . Let's assume the road is the line AB . The angle $\angle OAB$ is the angle between the road and the line AO . The angle $\angle OBA$ is the angle between the road and the line BO . Let's

assume the road is 090°T . Then $\angle OAB = 90 - 53 = 37^\circ$. $\angle OBA = 342 - 270 = 72^\circ$. Then $\angle AOB = 180 - (37 + 72) = 71^\circ$. (i) $\angle AOB = 71^\circ$ (assuming road is East-West). (ii) In $\triangle OTA$ (right-angled at O):

$$\tan 21^\circ = \frac{h}{OA} \implies OA = h \cot 21^\circ$$

In $\triangle OTB$ (right-angled at O):

$$\tan 26^\circ = \frac{h}{OB} \implies OB = h \cot 26^\circ$$

Using the Sine Rule in $\triangle AOB$:

$$\frac{OA}{\sin 72^\circ} = \frac{230}{\sin 71^\circ} \implies OA = \frac{230 \sin 72^\circ}{\sin 71^\circ} \approx 232.3$$

Now find h :

$$\tan 21^\circ = \frac{h}{232.3} \implies h = 232.3 \tan 21^\circ \approx 87 \text{ m}$$

Question 7:

Let OT be the tower of height h . B is due south of O ($\angle AOB = 180^\circ\text{T}$). C is on a bearing of 120°T from O . Thus, $\angle BOC = 120^\circ - 180^\circ = -60^\circ$, so $\angle BOC = 60^\circ$. In $\triangle OTB$ (right-angled at O):

$$\tan 23^\circ = \frac{h}{OB} \implies OB = h \cot 23^\circ$$

In $\triangle OTC$ (right-angled at O):

$$\tan 18^\circ = \frac{h}{OC} \implies OC = h \cot 18^\circ$$

In $\triangle OBC$ (on the ground):

$$\begin{aligned} BC^2 &= OB^2 + OC^2 - 2(OB)(OC) \cos 60^\circ \\ 200^2 &= (h \cot 23^\circ)^2 + (h \cot 18^\circ)^2 - 2(h \cot 23^\circ)(h \cot 18^\circ)(0.5) \\ 40000 &= h^2(\cot^2 23^\circ + \cot^2 18^\circ - \cot 23^\circ \cot 18^\circ) \\ \cot 23^\circ &\approx 2.3559, \quad \cot 18^\circ \approx 3.0777 \\ 40000 &= h^2(5.550 + 9.472 - 7.249) \\ 40000 &= h^2(7.773) \\ h^2 &= 5146.3 \implies h \approx 71.7 \text{ m} \end{aligned}$$

Question 8:

Let h be the height of the tree. X is due south of the tree Z , and Y is due east of the tree Z . Thus, $\angle XZY = 90^\circ$. (i) In $\triangle TZX$ (right-angled at Z):

$$\tan 20^\circ = \frac{h}{XZ} \implies XZ = h \cot 20^\circ$$

In $\triangle TZY$ (right-angled at Z):

Question 9:

Let BD be the flagpole of height h . A is 12 m from B . The cable AD is inclined at 30° to the horizontal. In $\triangle ABD$ (right-angled at B):

$$\tan 30^\circ = \frac{h}{12} \implies h = 12 \tan 30^\circ = 12 \times \frac{1}{\sqrt{3}} = 4\sqrt{3} \text{ m}$$

(a) Point C is 7 m from A on a bearing of 210° from A . The line AB is the direction of the cable, but we need the bearing of B from A . If the cable AD is tied to the floor 12 m from B , and B is the foot of the pole, the bearing of B from A is not given. Let's assume B is due North of A . Then the bearing of B from A is 000° . The bearing of C from A is 210° . Then $\angle BAC = 210^\circ - 0^\circ = 210^\circ$ (reflex). The interior angle $\angle BAC = 360 - 210 = 150^\circ$. This doesn't match 60° . Let's assume B is due South of A . Then the bearing of B from A is 180° . The bearing of C from A is 210° . Then $\angle BAC = 210 - 180 = 30^\circ$. Still not 60° . Let's assume the bearing of B from A is 150° . Then $\angle BAC = 210 - 150 = 60^\circ$. This is the most likely intended setup.

$$\angle BAC = 60^\circ$$

(b) The height BD was found in the first step: $BD = 4\sqrt{3}$ m. (c) To find BC , use the Cosine Rule in $\triangle ABC$ with $AB = 12$, $AC = 7$, $\angle BAC = 60^\circ$:

$$\begin{aligned} BC^2 &= 12^2 + 7^2 - 2(12)(7) \cos 60^\circ \\ BC^2 &= 144 + 49 - 168(0.5) \\ BC^2 &= 193 - 84 = 109 \\ BC &= \sqrt{109} \text{ m} \end{aligned}$$

Now find θ (the angle of elevation of cable CD):

$$\begin{aligned} \tan \theta &= \frac{BD}{BC} = \frac{4\sqrt{3}}{\sqrt{109}} \\ \theta &= \tan^{-1} \left(\frac{4\sqrt{3}}{\sqrt{109}} \right) \approx 24^\circ \end{aligned}$$